BUSINESS SCHOOL

SKEMA

# Corporate Finance
PGE M1 Fall 2026

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## Session 6: Bonds

# Housekeeping

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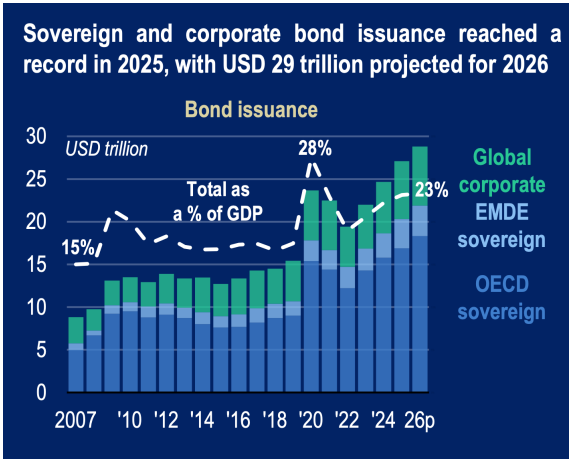
# Bonds

A bond is a **debt security** in which the issuer/seller (borrower) raises money from investors today in exchange for a promised repayment at the maturity date.

Example. In July 2026, Amazon raised $25bn through a bond sale to finance its capital expenditures in AI investments. Terms include:

- ▷ Maturities ranging from 3 to 40 years
- ▷ At 65 to 145 basis points (.65-1.45%) above U.S. treasuries
- ⇔ Coupon rates between $4.6 - 6.25\%$

# Some Facts: Corporate vs Govt Bonds

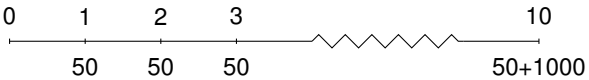



Source: Global Debt Report 2026, *OECD*

# Bond Terminology

Bonds typically pay periodic **coupon payments** as a source of intermediate cash flows for investors. At maturity, the issuer repays the principal, or the **face value** of the bond.

Example. Suppose the U.S. issues a 10-year treasury note with a face value of $1000 and a coupon rate of 5%. Then, investors will receive




- !! The coupon rate $\neq$ the interest rate on the bond (but is like an interest payment).

Corporations. Investment-grade, junk, callable, convertible bonds.
Governments. Treasury bills, notes, bonds, municipal bonds (local)

# Zero-Coupon Bond

A **zero-coupon** bond is a bond that only repays the face value at maturity (there are no coupon payments).

Example. Suppose the U.S. issues a 5-year a zero-coupon bond with face value of \$1000. If investors are willing to pay \$783.53 for this bond, what is the implied rate of interest?

$$\text{price} = \frac{\text{face value}}{(1+\text{rate})^t}$$

$$783.53 = \frac{1000}{(1+r)^5}$$

$$r = \left(\frac{1000}{783.53}\right)^{1/5} - 1 = 5\%$$

How to interpret? An investor pays \$783.53 today to receive \$1000 in 5 years. This gives a return of (1000-783.53)/783.53=27.6%, which, on an annual basis, is 5%.

# Coupon Bond



A **coupon-bearing** bond is a bond that pays the holder periodic interest payments (called coupons), paid typically semi-annually or annually. The value of a coupon payment is given by:

$$\text{Coupon} = \frac{\text{Coupon Rate} \times \text{Face Value}}{\text{Number of payments per year}}$$

- The coupon rate is always quoted as an annual rate

# Yield-to-Maturity (YTM)



The **yield-to-maturity** of a bond is the unique discount rate that sets the present value of all promised cash flow payments equal to the current market price.

Suppose the price today of a 2-year coupon bond with $F = 1000$, $C = 60$ is $p = 981.92$. Then,

$$p = 981.92 = \frac{60}{1+y} + \frac{1060}{(1+y)^2} \quad \Rightarrow y = 7\%$$

# Some info re: YTM

i The YTM is the average rate of return of purchasing a bond today and **holding the bond to maturity** while reinvesting intermediate coupon payments.

ii Memorize the below

| When the bond price is | We say the bond trades | This occurs when |
|---|---|---|
| greater than the face value | "above par" or "at a premium" | Coupon Rate $>$ Yield to Maturity |
| equal to the face value | "at par" | Coupon Rate $=$ Yield to Maturity |
| less than the face value | "below par" or "at a discount" | Coupon Rate $<$ Yield to Maturity |

# Some info re: YTM

ii (Continued)

- Due to the time-value of money, a zero-coupon must always trade at a discount.
- A coupon-bearing bond trades at “par” if the coupon rate = yield-to-maturity of that bond.

# Some info re: YTM

iii Mechanically, there is an inverse relationship between the bond price and its yield

# Exercise 1

Suppose a seven-year, $1000 bond with an 8% coupon rate and semi-annual coupons is trading with a yield to maturity of 6.75%.

a. Is this bond currently trading at a discount, at par, or at a premium? Explain.

# Exercise 1

Suppose a seven-year, $1000 bond with an 8% coupon rate and semi-annual coupons is trading with a yield to maturity of 6.75%.

b. If the yield to maturity of the bond rises to 7% (APR* with semi-annual compounding), what price will the bond trade for?

# Exercise 1

Suppose a seven-year, $1000 bond with an 8% coupon rate and semi-annual coupons is trading with a yield to maturity of 6.75%.

b. If the yield to maturity of the bond rises to 7% (APR* with semi-annual compounding), what price will the bond trade for?

$$p = \underbrace{\frac{C}{r}\left[1 - \left(\frac{1}{1+r}\right)^{T}\right]}_{\text{PV of coupon payments}} + \underbrace{\frac{F}{(1+r)^{T}}}_{\text{PV of face value}}$$

$$p = \frac{40}{.035}\left[1 - \left(\frac{1}{1.035}\right)^{14}\right] + \frac{1000}{(1.035)^{14}}$$

$$p = 1054.60$$

# Interest Rate Risk: Maturity



- Suppose we have a 5-year ZCB with a face value of 100 and a YTM=5%. What happens if the yield-to-maturity increases by one percentage point to 6%?

$$\frac{100}{1.05^5} = 78.35 \quad \rightarrow \quad \frac{100}{1.06^5} = 74.73$$

Prices fell by 3.63, which is a percentage drop of (74.73-78.35)/78.35= -4.6%.

- What if we were holding a 10-year ZCB?

$$\frac{100}{1.05^{10}} = 61.39 \quad \rightarrow \quad \frac{100}{1.06^{10}} = 55.84$$

which is a percentage drop of (55.84-61.39)/61.39= -9%.

# Interest Rate Risk: Coupon Rates



- Suppose we have a 2-year coupon with a face value of 100, a coupon rate of 2% and a YTM=5%. What happens if the yield-to-maturity increases by one percentage point to 6%?

$$\frac{2}{1.05} + \frac{102}{1.05^2} = 94.42 \quad \rightarrow \quad \frac{2}{1.06} + \frac{102}{1.06^2} = 92.67$$

which is a percentage drop of (92.67-94.42)/94.42= -1.9%.

- What if the coupon rate was 5%?

$$100 \quad \rightarrow \quad \frac{5}{1.06} + \frac{105}{1.06^2} = 98.17$$

which is a percentage drop of (98.17-100)/100= -1.8%.

# Drivers of Interest Rate Risk

All else equal,

- **Maturity**
  Longer maturity $\rightarrow$ higher interest rate risk

- **Coupon Rate**
  Higher coupon rates $\rightarrow$ lower interest rate risk

# Exercise 2

Consider the following bonds:

| Bond | Coupon Rate | Maturity (years) |
|---|---|---|
| A | 0% | 15 |
| B | 0% | 10 |
| C | 4% | 15 |
| D | 8% | 10 |

a. What is the percentage change in the price of each bond if its yield to maturity falls from 6% to 5%?

# Exercise 2

| Bond | Coupon Rate (annual payments) | Maturity (years) | Price at 6% YTM | Price at 5% YTM | Percentage Change |
|---|---|---|---|---|---|
| A | 0% | 15 | $41.73 | $48.10 | 15.3% |
| B | 0% | 10 | $55.84 | $61.39 | 9.9% |
| C | 4% | 15 | $80.58 | $89.62 | 11.2% |
| D | 8% | 10 | $114.72 | $123.17 | 7.4% |

# Exercise 2

Consider the following bonds:

| Bond | Coupon Rate | Maturity (years) |
|---|---|---|
| A | 0% | 15 |
| B | 0% | 10 |
| C | 4% | 15 |
| D | 8% | 10 |

b. Which of the bonds A–D is most sensitive to a 1% drop in interest rates from 6% to 5% and why? Which bond is least sensitive? Provide an intuitive explanation for your answer.

# Credit Rating Scale

| | Moody's | S&P / Fitch | Risk-free + |
|---|---|---|---|
| Investment Grade | Aaa | AAA | 0.3%–0.6% |
| | Aa1–Aa3 | AA+/AA/AA- | 0.5%–0.9% |
| | A1–A3 | A+/A/A- | 0.7%–1.3% |
| | Baa1–Baa3 | BBB+/BBB/BBB- | 1.2%–2.2% |
| Speculative Grade | Ba1–Ba3 | BB+/BB/BB- | 2.5%–4.5% |
| | B1–B3 | B+/B/B- | 4.5%–7% |
| | Caa1–Caa3 | CCC+/CCC/CCC- | 7%–12%+ |
| | Ca | CC/C | 12%+ (distressed) |
| | C/D | D | N/A |

# Global corporate average default rates (1981–2025) (%)

| | Time horizon (years) | |
|---|---|---|
| Rating | 5 | 10 |
| AAA | 0.34 | 0.68 |
| AA | 0.27 | 0.65 |
| A | 0.38 | 0.99 |
| BBB | 1.33 | 2.78 |
| BB | 5.59 | 10.10 |
| B | 15.34 | 21.67 |
| CCC/C | 46.24 | 50.00 |
| Investment-grade | 0.76 | 1.65 |
| Speculative-grade | 13.49 | 18.87 |

* Source: S&P Global.

# Exercise 3

The following table summarizes the yields to maturity on several one-year, zero-coupon securities:

| Security | Yield (%) |
|---|---|
| Treasury | 3.1 |
| AAA corporate | 3.2 |
| BBB corporate | 4.2 |
| B corporate | 4.9 |

a. What is the price (expressed as a percentage of the face value) of a one-year, zero-coupon corporate bond with a AAA rating ?

# Exercise 3

The following table summarizes the yields to maturity on several one-year, zero-coupon securities:

| Security | Yield (%) |
|---|---|
| Treasury | 3.1 |
| AAA corporate | 3.2 |
| BBB corporate | 4.2 |
| B corporate | 4.9 |

b. What is the credit spread on AAA-rated corporate bonds?

c. What is the credit spread on B-rated corporate bonds?

# Exercise 3

The following table summarizes the yields to maturity on several one-year, zero-coupon securities:

| Security | Yield (%) |
|---|---|
| Treasury | 3.1 |
| AAA corporate | 3.2 |
| BBB corporate | 4.2 |
| B corporate | 4.9 |

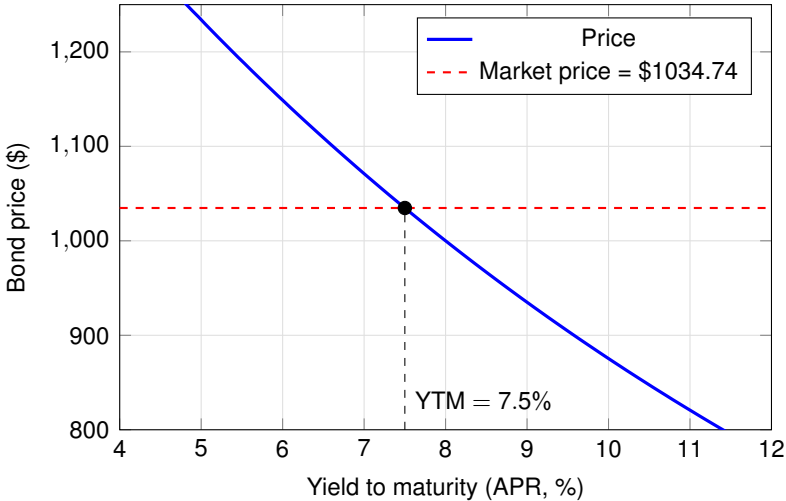
d. How does the credit spread change with the bond rating? Why?

# Exercise 4

Suppose a 10-year, $1000 bond with an 8% coupon rate and semiannual coupons is trading for a price of $1034.74.

a. Determine graphically, what is the bond's yield to maturity

# Bond price vs. yield to maturity



# Exercise 4



Suppose a 10-year, $1000 bond with an 8% coupon rate and semiannual coupons is trading for a price of $1034.74.

b. If the bond's yield to maturity changes to 9% APR, what will the bond's price be ?
   $p = 934.96$

# Exercise 4

Suppose a 10-year, $1000 bond with an 8% coupon rate and semiannual coupons is trading for a price of $1034.74.

c. At 9% what is the sensitivity of the bond? Explain this result. And the duration ?

# Duration

The **duration** of a bond is a measure of its sensitivity to interest rate changes. Quite literally, it measures how many years it takes for a bond to pay back its face value.

Let $p$ denote the price of a bond given maturity $N$ and YTM $y$. The **Macaulay duration** is given by

$$D_{\text{Macaulay}} = \frac{1}{p} \sum_{n=1}^{N} n \times \frac{C_n}{(1+y)^n}$$

where $C_n$ are all cash flows at time $n$ (including the face value).

# Duration for a ZCB



# Duration for a ZCB

For a single cash flow $F$ received in $N$ periods, the price is $P = \dfrac{F}{(1+y)^N}$. Differentiating with the chain rule:

$$\frac{dP}{dy} = -N\,F\,(1+y)^{-N-1} = -\frac{N}{1+y} \cdot \frac{F}{(1+y)^N} = -\frac{N}{1+y}\,P$$

So the relative sensitivity is

$$\frac{1}{P}\frac{dP}{dy} = -\frac{1}{1+y} \times N$$

The Macaulay duration of this bond is $N$.

# Macaulay Duration

Based on

$$D_{\text{Macaulay}} = \frac{1}{p} \sum_{n=1}^{N} n \times \frac{C_n}{(1+y)^n}$$

We can construct a table for a bond with a coupon rate of 5%, face value of 100 and YTM of 6%.

| Time (years) | Cash flow | Present value | Weight | Time $\times$ weight |
|---|---|---|---|---|
| 1 | 5 | 4.717 | 0.049 | 0.049 |
| 2 | 5 | 4.450 | 0.046 | 0.093 |
| 3 | 5 | 4.198 | 0.044 | 0.131 |
| 4 | 5 | 3.960 | 0.041 | 0.165 |
| 5 | 105 | 78.463 | 0.819 | 4.097 |
| *Duration* | 125 | 95.787 | 1.000 | 4.535 |

# Modified Duration

The **modified duration** is given by

$$D_{\text{modified}} = \frac{1}{1+y} \times D_{\text{Macaulay}}$$

**Interpretation.** For a bond with a modified duration of $N$ years, a one

percentage point change in the yield leads to approximately $N$ percentage price change in the bond price.

# Modified Duration

Recall, a bond with a coupon rate of 5%, face value of 100 and YTM of 6% has a modified duration of 4.465 years and a price of 95.787 today. A one percentage point *decrease* in the YTM from 6% to 5% leads to...

Since the coupon rate = YTM, the bond trades at par and $p = 100$. The percentage price change is

$$\frac{\Delta p}{p} = \frac{100 - 95.787}{95.787} = 4.4\%$$

For comparison,

$$D_{\text{modified}} = \frac{1}{1+y} \times D_{\text{Macaulay}} = \frac{1}{1.06} \times 4.535 = 4.28\%$$

so the interpretation is correct, though duration slightly underestimates the price increase when yields fall.

# First and Second Order Terms (Extra)

Let $p$ denote the price of a bond, and let $p(y)$ indicate that $p$ is a function of $y$. For a small change in the ytm,

$$p(y+\Delta y) \approx p(y) + p'(y)\Delta y + \frac{1}{2}p''(y)(\Delta y)^2 + \cdots$$

$$\Delta p \approx p'(y)\Delta y + \frac{1}{2}p''(y)(\Delta y)^2 + \cdots$$

$$\frac{\Delta p}{p} \approx \frac{p'(y)}{p}\Delta y + \frac{1}{2}\frac{p''(y)}{p}(\Delta y)^2 + \cdots$$

$$\frac{\Delta p}{p} \approx -D_{\text{mod}}\Delta y + \frac{1}{2}\frac{p''(y)}{p}(\Delta y)^2 + \cdots$$

When $\Delta y$ is small, the second-order terms $\to 0$